Churn Analysis & Customer Intelligence

OTT subscription platform | Python + SQL



Tools: Python, SQL (SQLite), pandas, NumPy, matplotlib, seaborn

Business Problem & Approach

The problem

In the competitive OTT market (Netflix, Hotstar, Prime), retention drives revenue. The goal: find which subscribers leave, why they leave, and when, then recommend how to keep them.

Data (3 related tables)

db_customer - demographics, state, interests

db_subscription - plan, contract, charges, CLTV, churn score

db_support - complaints, escalations, CSAT

Approach

1. Pulled multi-table data from SQL into Python (sqlite3 + pandas)
2. Cleaned data: types, nulls, column selection, QC checks
3. Engineered features: tenure, churn flag, customer age
4. EDA: group-by and pivot analysis across plan, contract, state, time
5. Visualised patterns with matplotlib and seaborn
6. Converted findings into business recommendations

Key Metrics

28.6%
Churn rate

71.4%
Retention rate

1,451
Avg tenure (days)

18.8
ARPU

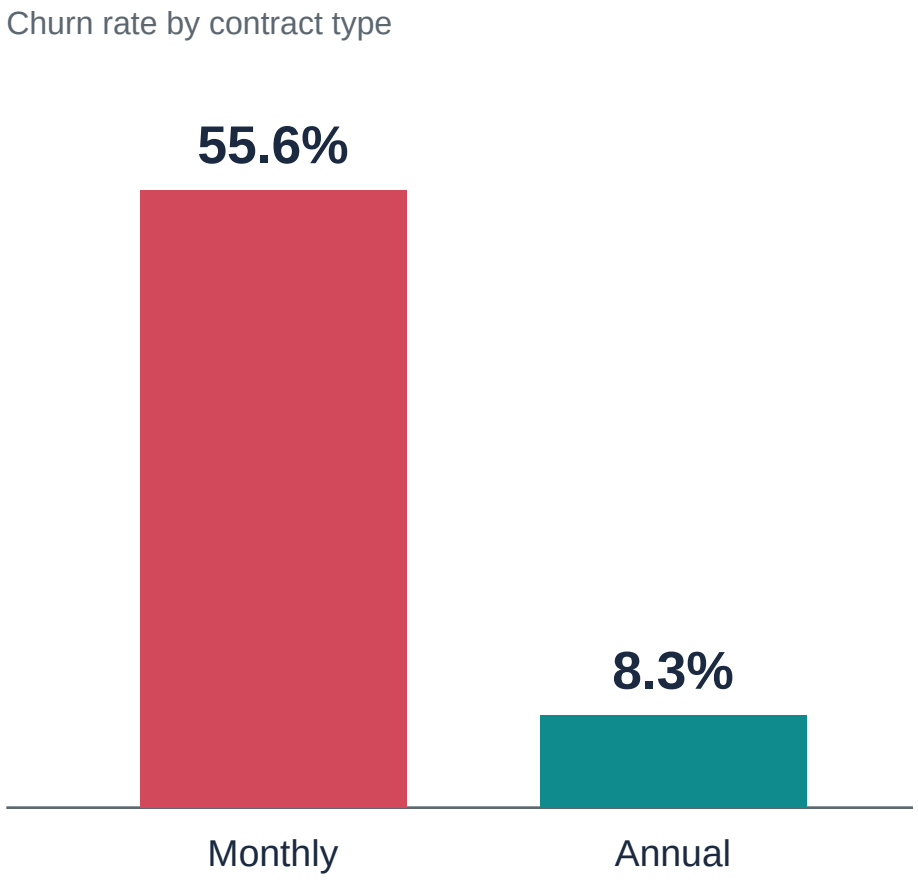
395
Total revenue

74
Revenue lost to churn (18%)

2,047
CLTV lost

6.7x
Monthly vs annual churn gap

Key Insight: Contract Type Drives Churn



- Monthly-contract subscribers churn **6.7x more** than annual (55.6% vs 8.3%).
- Most churn comes from the **Basic plan**, with limited revenue impact.
- Churn peaked in **September 2024**; **Karnataka** is the most affected state.
- Churn cost **74 in revenue (18%)** and **2,047 in CLTV**.

Recommendations

1 **Move monthly subscribers to annual contracts**

Offer an annual-plan incentive to the highest-churn segment.

2 **Investigate Karnataka and September 2024**

Check for price changes, complaint spikes or technical issues in that state and month.

3 **Prioritise High and Medium churn-risk customers**

Rank by CLTV, review their complaints, and reach out by email, SMS or call.

4 **Review the Basic plan and competitor activity**

Check recent pricing changes and why some customers switched to a competitor.